

Title: Hybrid RAG Framework for Enterprise Knowledge Management

Abstract:

Organizations generate and maintain a vast amount of enterprise knowledge in the form of annual reports, policy documents, technical manuals, product documentation, financial reports, and internal knowledge bases. Retrieving relevant information from these large collections of documents is often time-consuming and inefficient using conventional keyword-based search systems. Although Retrieval-Augmented Generation (RAG) has significantly improved enterprise question-answering by combining document retrieval with Large Language Models (LLMs), traditional RAG frameworks primarily rely on semantic retrieval, which may fail to identify exact keyword matches or domain-specific information, resulting in incomplete or less relevant responses.

This project proposes a **Hybrid RAG Framework for Enterprise Knowledge Management** that integrates **semantic retrieval** and **keyword-based retrieval** to improve the accuracy and relevance of enterprise information retrieval. The proposed framework utilizes **LangChain** for workflow orchestration, **Sentence Transformers** for generating semantic embeddings, **FAISS** for efficient vector similarity search, **BM25** for keyword-based retrieval, and **Llama 3** as the Large Language Model for generating context-aware responses. Enterprise documents from publicly available company reports, technical documentation, and corporate publications are preprocessed, segmented into meaningful text chunks, embedded into a vector database, and retrieved using a hybrid retrieval mechanism that combines both semantic similarity and lexical matching.

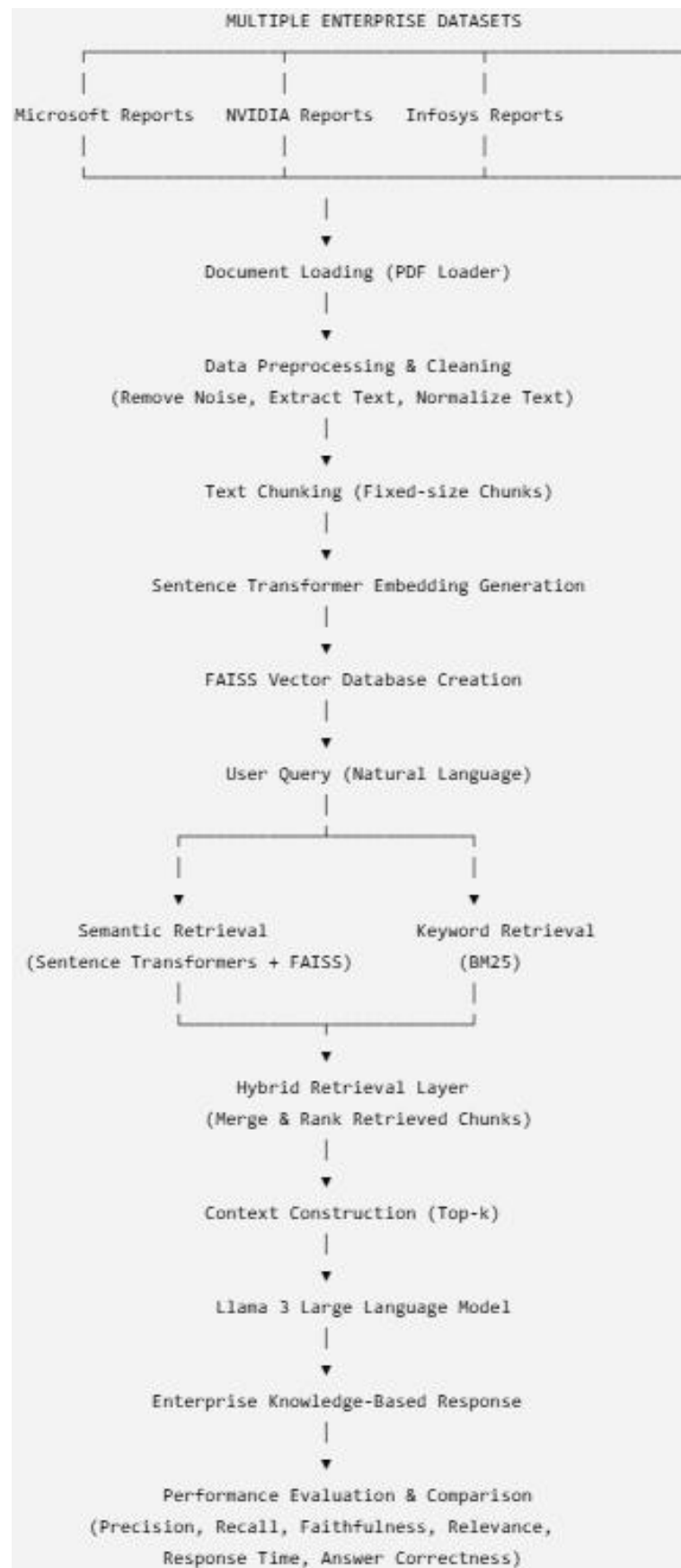
The proposed system enables users to interact with enterprise documents using natural language queries and provides accurate, context-aware answers by retrieving the most relevant information before response generation. The framework aims to reduce hallucinations, improve retrieval precision, and enhance the reliability of enterprise knowledge management systems. The performance of the proposed model will be evaluated using multiple enterprise document collections and compared with a

standard RAG-based retrieval model using evaluation metrics such as **retrieval precision, recall, answer correctness, answer relevance, faithfulness, and response time**. The resulting system is expected to provide an efficient, scalable, and intelligent enterprise knowledge assistant that improves information accessibility and supports faster organizational decision-making.

Dataset:

Microsoft Annual Reports & Investor Documents	Annual reports, 10-K filings, shareholder letters, and enterprise business information.	https://www.microsoft.com/en-us/investor/annual-reports?msocid=2544b1e5d29b620414cda763d3e263a7&utm_source=chatgpt.com
NVIDIA Annual Reports	Annual reports covering AI products, business strategy, financials, and technology, ESG & Sustainability Reports	https://investor.nvidia.com/financial-info/annual-reports-and-proxies/default.aspx?utm_source=chatgpt.com
Infosys Annual Reports	Annual reports, Form 20-F filings, and corporate reports.	https://www.infosys.com/investors/reports-filings/annual-report/annual-reports.html?utm_source=chatgpt.com

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